higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

STRENGHT OF MATERIALS AND STRUCTURES N6

2 DECEMBER 2019

This marking guideline consists of 8 pages.

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QUESTION 1

1.1 $$at\ \ 100\ \ mm\ :a-\frac{b}{0,1^2}=46\times10^6\ldots\ldots\ldots(1)$$

$$at\ \ 100\ \ mm\ :a+\frac{b}{0,1^2}=30\times10^6\ldots.(2)$$

$$(1)+(2)\ :2a=76\times10^6$$

$$a=38\times10^6 \qquad b=-0,08\times10^6$$

$$at\ \ 200\ \ mm\ :\ \sigma_H=a-\frac{b}{0,2^2}$$

$$=38\times10^6-\frac{-0,08\times10^6}{0,2^2}$$

$$\sigma_H=40\ \ MPa\ \ (compressive)$$

$$at\ \ 200\ \ mm\ :\ \sigma_R=a+\frac{b}{0,2^2}$$

$$=38\times10^6+\frac{-0,08\times10^6}{0,2^2}$$

$$\sigma_R=36\ \ MPa\ \ (compressive)$$ (6)

1.2 $$at\ \ 200\ \ mm\ :\quad a+\frac{b}{0,2^2}=36\times10^6\ldots\ldots\ldots(3)$$

$$at\ \ 400\ \ mm\ :a+\frac{b}{0,4^2}=0\ldots\ldots\ldots\ldots\ldots\ldots\ldots.(4)$$

$$(3)-(4)\ :25b-6,25b=36\times10^6$$

$$b=1,92\times10^6$$
$$a=-12\times10^6$$

$$at\ \ 200\ \ mm\ :\ \sigma_H=a-\frac{b}{0,2^2}$$

$$=-12\times10^6-\frac{1,92\times10^6}{0,2^2}$$

$$\sigma_H=-60\ \ MPa\ \ (tensile)$$

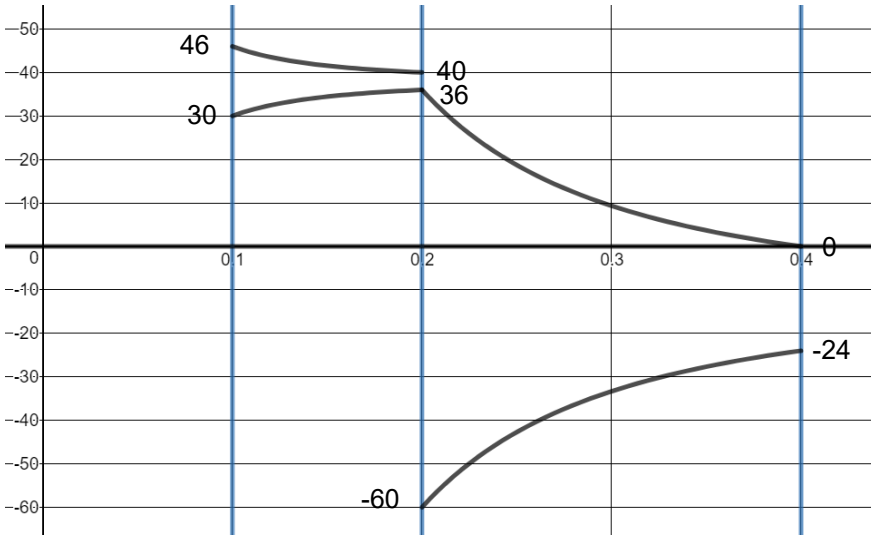
$$at\ \ 400\ \ mm\ :\ \sigma_H=a-\frac{b}{0,4^2}$$ (6)

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$$= -12 \times 10^6 - \frac{1,92 \times 10^6}{0,4^2}$$

$$\sigma_H = -24 \ \ MPa \ \ (tensile)$$

1.3



(4)

[16]

QUESTION 2

2.1

$$y_1 = \frac{w l_1^4}{8 EI} + \frac{w l_1^3 \times l_2}{6 EI}$$

$$= \frac{20 \times 10^3 \times 1,25^4}{8 \times 200 \times 10^9 \times I} + \frac{20 \times 10^3 \times 1,25^3 \times 1,25}{6 \times 200 \times 10^9 \times I}$$

$$y_1 = \frac{71,208 \times 10^{-9}}{I}$$

$$y_2 = \frac{F L^3}{3 EI} = \frac{2 \times 10^3 \times 2,5^3}{3 \times 200 \times 10^9 \times I} = \frac{52,083 \times 10^{-9}}{I}$$

$$y_T = y_1 + y_2$$

$$7 \times 10^{-3} = \frac{71,208 \times 10^{-9}}{I} + \frac{52,083 \times 10^{-9}}{I}$$

$$I = 17,613 \times 10^{-6} m^4$$

$$I = \frac{\pi \left(D^4 - d^4 \right)}{64}$$

$$17,613 \times 10^{-6} = \frac{\pi \left((2d)^4 - d^4 \right)}{64}$$

$$d = 70 \ \ mm$$
$$D = 140 \ \ mm$$

(8)

2.2 $For \;\; I = 17{,}613 \times 10^{-6}\, m^4 \;\; select \;\; 203 \times 102 \times 25{,}3 \;\; kg/m$ (1)

2.3
$$M = FL + \frac{w\,l^2}{2}$$
$$= 2 \times 10^3 \times 2{,}5 + \frac{20 \times 10^3 \times 1{,}25^2}{2}$$
$$M = 20{,}625 \;\; kNm$$
$$\sigma = \frac{M}{Z} = \frac{20{,}625 \times 10^3}{226{,}1 \times 10^{-6}} = 91{,}221 \;\; MPa$$
(3)
[12]

QUESTION 3

3.1
$$A = \pi r^2 = \pi \times 0{,}015^2 = 706{,}858 \times 10^{-6} \;\; m^2$$
$$\sigma_d = \frac{F}{A} = \frac{F}{706{,}858 \times 10^{-6}} = 1414{,}711\, F$$
$$I = \frac{\pi d^4}{64} = \frac{\pi \times 0{,}03^4}{64} = 39{,}761 \times 10^{-9} \;\; m^4$$
$$\sigma_b = \frac{F \times e \times Y}{I} = \frac{F \times 0{,}09 \times 0{,}015}{39{,}761 \times 10^{-9}} = 33953{,}055\, F$$
$$\sigma_{Rmax} = \sigma_d + \sigma_b$$
$$90 \times 10^6 = 1414{,}711\, F \;\; + 33953{,}055\, F$$
$$F = 2544{,}69 \;\; N \;\; \leftrightarrow m = 259{,}4 \;\; kg$$
(7)

3.2
$$\sigma_d = 1414{,}711\, F = 3{,}6 \;\; MPa \;\; (tensile)$$
$$\sigma_b = 33953{,}055\, F = 86{,}4 \;\; MPa$$
$$\sigma_{min} = \sigma_b - \sigma_d = 82{,}8 \;\; MPa \;\; (compressive)$$
(4)
[11]

QUESTION 4

4.1 $W = \rho g A l = 2500 \times 9,81 \times 2,1 \times h \times 1 = 51502,5\,h$

$$F_w = \frac{\rho g\, h^2}{2} = \frac{1000 \times 9,81 \times h^2}{2} = 4905\,h^2$$

$$Moments\ \ about\ \ the\ \ toe: \quad V \times x_R + F_g \times \frac{h}{3} = W \times x_1$$

$$51502,5\,h \times 0,6 \ + \ 4905\,h^2 \times 0,333\,h \ = 51502,5\,h \times 1,05$$

$$h = 3,765 \ \ m$$ (6)

4.2 $W \sim M = W \times x_1 = (51502,5 \times 3,765) \times 1,05 = 203,601 \ \ kNm$

$$F \sim M = F_w \times \frac{h}{3} = 4905\,h^2 \times \frac{h}{3} = 4905 \times 3,765^2 \times \frac{3,765}{3} = 87,257 \ \ kNm$$

$$FOS = \frac{W \sim M}{F \sim M} = \frac{203,601}{87,257} = 2,3 \quad (safe) > 2$$ (3)

4.3 $e = 0,5\,B - x = 1,05 - 0,6 = 0,45 \ \ m$

$$V = W = 51502,5\,h = 51502,5 \times 3,765 = 193,905 \ \ kN$$

$$\sigma_{max} = \frac{V}{B} + \frac{6\,Ve}{B^2} = \frac{193,905}{2,1} + \frac{6 \times 193,905 \times 0,45}{2,1^2} = 211,053 \ \ kPa \ \ (comp)$$

$$\sigma_{min} = \frac{V}{B} - \frac{6\,Ve}{B^2} = \frac{193,905}{2,1} - \frac{6 \times 193,905 \times 0,45}{2,1^2} = -26,382 \ \ kPa \ \ (tensile)$$ (4)

[13]

5.1 $W_c = 137 \times 9,81 \times 4 = 5,376 \ \ kN$

$$F_c = \sigma A = 9 \times 10^6 \times 17,42 \times 10^{-3} = 156,78 \ \ kN$$

$$W_T = F_c + W_c + W_f = 156,78 + 5,376 + 25 = 187,156 \ \ kN$$

$$p = \frac{W_T}{A} = \frac{187,156 \times 10^3}{1,2^2} = 129,969 \ \ kPa$$

$$\sigma_{ult} = p \times FOS = 129,969 \times 3 \ = 389,908 \ \ kPa$$ (6)

5.2 $$t = \frac{L - l}{\sqrt{2}} = \frac{1,2 - 0,5}{\sqrt{2}} \ = 0,495 \ \ m$$ (2)

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5.3 $$d=\frac{p}{\rho g}\left(\frac{1-\sin\theta}{1+\sin\theta}\right)^2=\frac{138,889\times10^3}{1600\times9,81}\left(\frac{1-\sin 30}{1+\sin 30}\right)^2=0,983\ \ m$$ (2)

[10]

QUESTION 6

6.1 $$\frac{\sigma_s}{\sigma_c}=\frac{m(d-n)}{n}$$

$$\frac{140}{8}=\frac{15(0,38-n)}{n}$$

$$n=0,175\ \ m$$ (2)

6.2 $$M_s=\sigma_s A_s(d-n)=140\times10^6\times6400\times10^{-6}(0,38-0,175)$$

$$M_s=183,335\ \ kNm$$

$$M_c=M-M_s=300-183,335=116,665\ \ kNm$$ (3)

6.3 $$\sigma_{c1}=\frac{\sigma_c(n-t)}{n}=\frac{8(0,175-0,1)}{0,175}=3,439\ \ MPa$$ (2)

6.4 $$M_c=\frac{1}{2}\sigma_c bn\frac{2}{3}n-\frac{1}{2}\sigma_{c1}(b-e)(n-t)\frac{2}{3}(n-t)$$

$$116665=\frac{8\times10^6\times b\times0,175^2}{3}-\frac{3,439\times10^6\times(b-0,27)(0,175-0,1)^2}{3}$$

$$116665=82026,0355\,b-6513,665\,b+1758,69$$

$$b=1,522\ \ m$$ (5)

[12]

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QUESTION 7

7.1 $w = \frac{3024}{2 \times 36} = 42 \ \ kN/m$ (1)

7.2 $F_H = \frac{w x^2}{2 d} = \frac{42 \times 10^3 \times 18^2}{2 \times 3} = 2268 \ \ kN$

$F_v = wx = 21 \times 10^3 \times 18 \ = 378 \ \ kN$

$F_T = \sqrt{F_H^2 + F_v^2} = \sqrt{2268^2 + 378^2} = 2299{,}284 \ \ kN$ (3)

7.3 $\sigma = \frac{\sigma_{ult}}{FOS} = \frac{320}{8} = 40 \ \ MPa$

$A = \frac{F}{\sigma} = \frac{2299{,}284 \times 10^3}{40 \times 10^6} = 57{,}482 \times 10^{-3} \ m^2$

$d = \sqrt{\frac{4A}{\pi}} = \sqrt{\frac{57{,}482 \times 10^{-3}}{\pi}} = 27{,}1 \ \ mm$ (3)

7.4 $At \ \ 10 \ \ m \ \ from \ \ lower \ \ support: \ \ x_2 = 18 - 10 = 8 \ \ m$

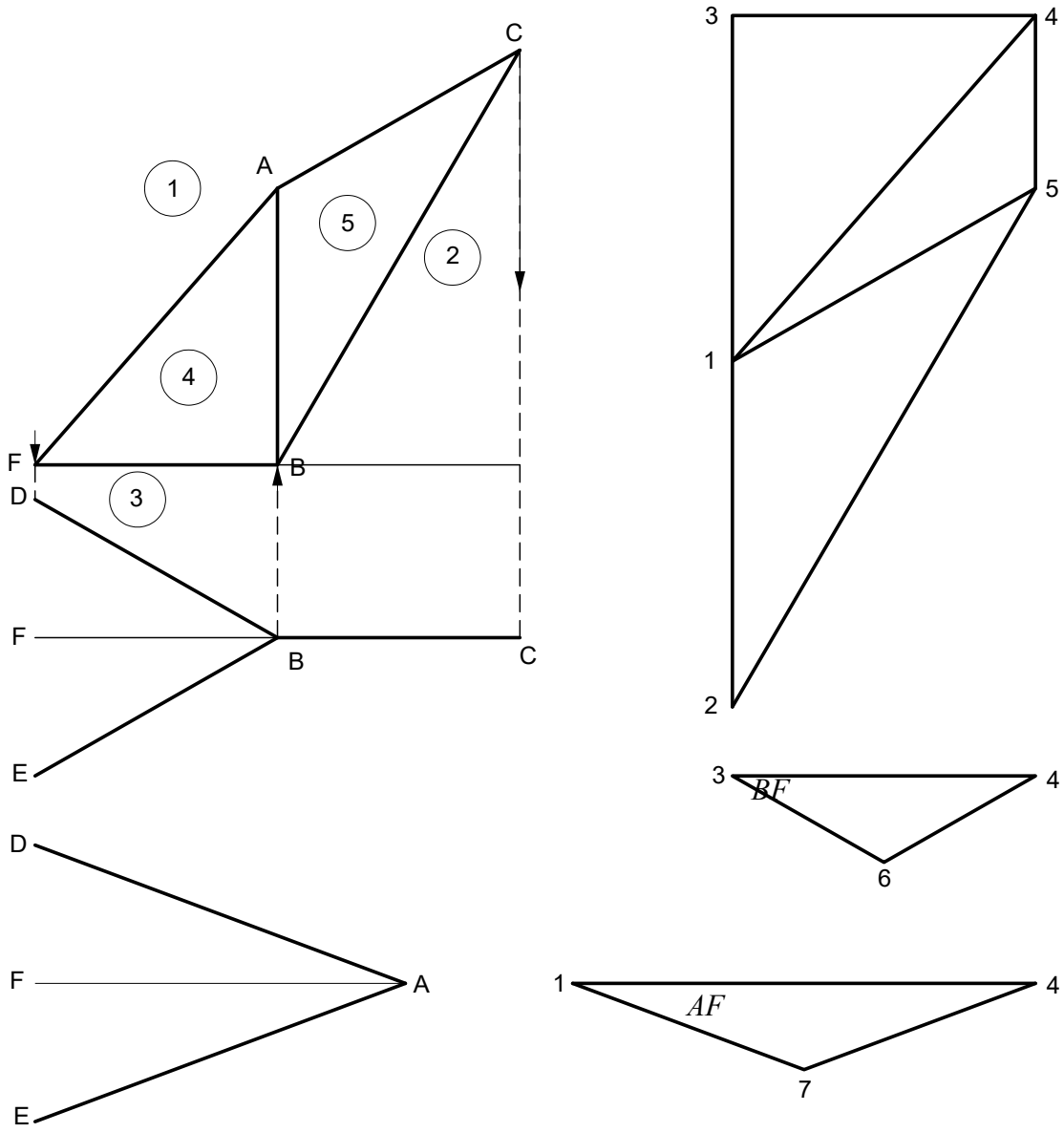
$F_{v2} = w x_2 = 21 \times 10^3 \times 8 = 168 \ \ kN$

$F_{T2} = \sqrt{F_H^2 + F_{v2}^2} = \sqrt{2268^2 + 168^2} = 2274{,}214 \ \ kN$ (3)

[10]

QUESTION 8

8.1 **SPACE DIAGRAMS** 8.1 and 8.2 **VECTOR DIAGRAMS**



(10)

8.3

MEMBER	MAGNITUDE	NATURE
AB (4-5)	25 kN √	Strut √
AC (1-5)	50 kN √	Tie √
BC (2-5)	86,6 kN √	Strut √
DB (3-6) and EB (6-4)	25 kN √	Strut √
DA (1-7) and EA (7-4)	35,4 kN √	Tie √
R_B (2-3)	100 kN √	Upwards√

(6)

$(NOTE: \sqrt{} = Half\ \ mark)$ **[16]**

TOTAL: 100

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